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Course Code: CBS3005

**Course Title: Cloud, Microservices and
Applications**

Class Number: VL2025260105679

Lab Assessment - 4

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Reg No: 22BBS0155

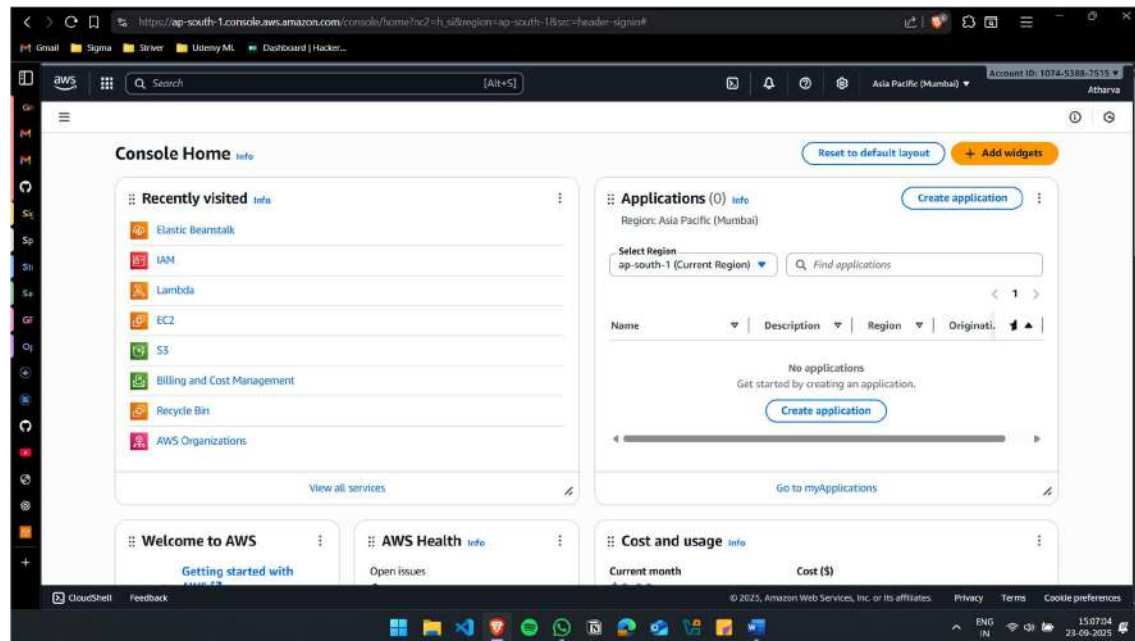
Q. A company wants to deploy a secure, scalable, and highly available web application on AWS for global users. Perform the following tasks in AWS and submit screenshots of each step as evidence:

- (i) Launch multiple EC2 instances (web servers) and configure them in different Availability Zones.
- (ii) Create an Application Load Balancer (ALB) to distribute traffic across these instances.
- (iii) Configure health checks so that faulty instances are automatically removed from load balancing.
- (iv) Enable Auto Scaling to add/remove instances based on traffic demand.
- (v) Configure path-based routing: /auth requests go to the authentication service, /order requests go to the order processing service. Integrate the Load Balancer with Route 53 so that global users are routed to the nearest AWS region.

➔ SOLUTION:

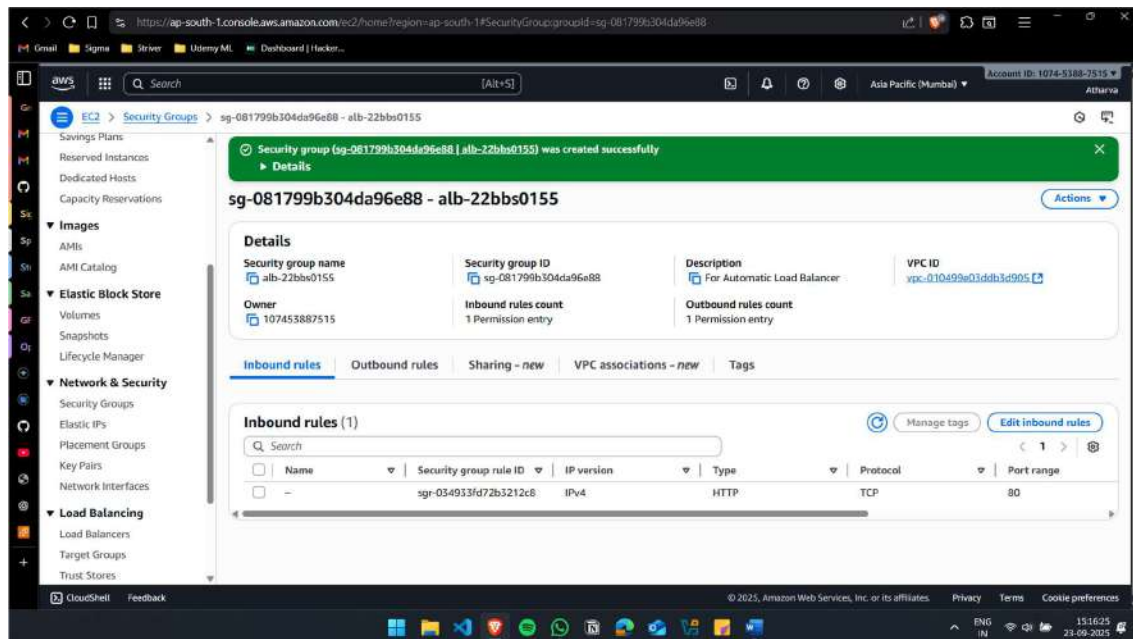
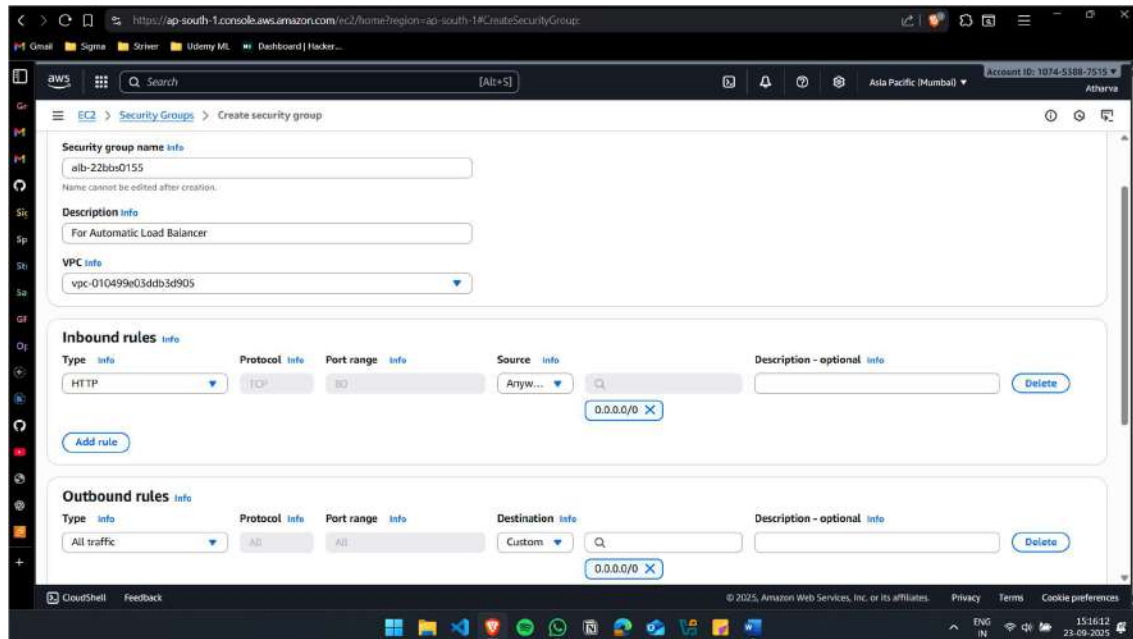
(i) Launch multiple EC2 instances (web servers) and configure them in different Availability Zones.

Initial:



Create 2 security group for the instances:

alb-22bbs0155: For Automatic Load Balancer



web-22bbs0155: For Web Server

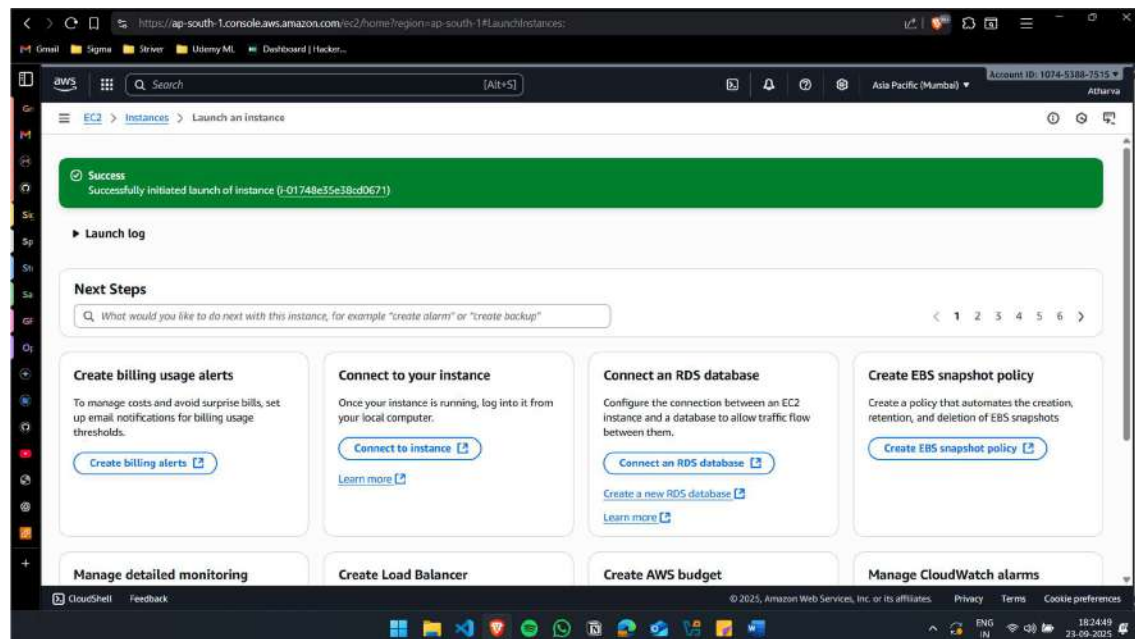
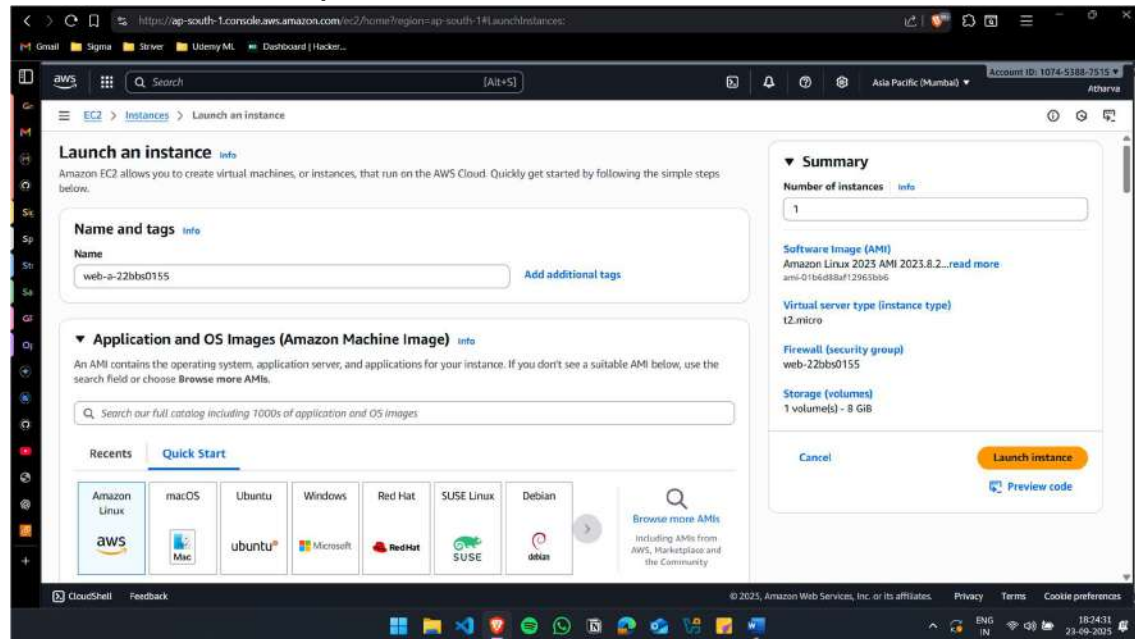
The screenshot shows the 'Create security group' page in the AWS Management Console. The page is for the 'web-22bbs0155' security group. The 'Security group name' field is filled with 'web-22bbs0155'. The 'Description' field is filled with 'For web server'. The 'VPC' dropdown is set to 'vpc-010499e03ddb5d905'. Under 'Inbound rules', a rule is added with 'Type' as 'HTTP', 'Protocol' as 'TCP', 'Port range' as '80', and 'Source' as 'Custom' with a search for 'sg-081799b304da96e' and a selection of 'sg-081799b304da96e88'. There is an 'Add rule' button. Under 'Outbound rules', a rule is added with 'Type' as 'All traffic', 'Protocol' as 'All', 'Port range' as 'All', and 'Destination' as 'Custom' with a search for '0.0.0.0/0' and a selection of '0.0.0.0/0'. There is a 'Delete' button. The bottom of the page shows the AWS logo, search bar, and navigation links.

The screenshot shows the 'Details' page for the security group 'sg-07f0318714957777d - web-22bbs0155'. A green banner at the top states 'Security group (sg-07f0318714957777d | web-22bbs0155) was created successfully'. The 'Details' section shows the 'Security group name' as 'web-22bbs0155', 'Security group ID' as 'sg-07f0318714957777d', 'Description' as 'For web server', 'Owner' as '107453887515', 'Inbound rules count' as '1 Permission entry', and 'Outbound rules count' as '1 Permission entry'. The 'VPC ID' is 'vpc-010499e03ddb5d905'. Below the details, there are tabs for 'Inbound rules', 'Outbound rules', 'Sharing - new', 'VPC associations - new', and 'Tags'. The 'Inbound rules (1)' tab is active, showing a table with one rule: 'sgr-0f9b03aa82f887082' with 'Type' as 'HTTP', 'Protocol' as 'TCP', and 'Port range' as '80'. The table has columns for 'Name', 'Security group rule ID', 'IP version', 'Type', 'Protocol', and 'Port range'. There are 'Manage tags' and 'Edit inbound rules' buttons. The bottom of the page shows the AWS logo, search bar, and navigation links.

The web server can have inbound only from the sg-alb

Create 2 EC2 Instances in different AZ

web-a-22bbs0155: AZ = ap-south-1a



web-b-22bbs0155: AZ = ap-south-1b

https://ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#instanceDetails:instanceId=i-0def76871d9736448

aws [Search] [Alt+S] Asia Pacific (Mumbai) Account ID: 1074-5188-7515

EC2 > Instances > i-0def76871d9736448

Instance summary for i-0def76871d9736448 (web-b-22bbs0155) info

Updated less than a minute ago

Instance ID
i-0def76871d9736448

IPv6 address
-

Public IPv4 address
13.203.156.195 | [open address](#)

Instance state
Running

Private IPv4 addresses
172.31.1.43

Public DNS
ec2-13-203-156-195.ap-south-1.compute.amazonaws.com | [open address](#)

Hostname type
IP name: ip-172-31-1-43.ap-south-1.compute.internal

Private IP DNS name (IPv4 only)
ip-172-31-1-43.ap-south-1.compute.internal

Answer private resource DNS name
-

Instance type
t2.micro

Auto-assigned IP address
13.203.156.195 [Public IP]

VPC ID
vpc-010499e03d905f905

IAM Role
-

Subnet ID
subnet-0e18ad7b4b3e53b14

Instance ARN
arn:aws:ec2:ap-south-1:107453887515:instance/i-0def76871d9736448

Managed
false

Operator
-

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https://ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#instances

aws [Search] [Alt+S] Asia Pacific (Mumbai) Account ID: 1074-5188-7515

EC2 > Instances

Instances (1/2) info

Last updated less than a minute ago

Connect Instance state Actions Launch instances

Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public
web-b-22bbs0...	i-0def76871d9736448	Running	t2.micro	Initializing	View alarms +	ap-south-1b	ec2-1
web-a-22bbs0...	i-01748e35e38cd0671	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	ec2-3

i-0def76871d9736448 (web-b-22bbs0155)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary info

Instance ID
i-0def76871d9736448

IPv6 address
-

Public IPv4 address
13.203.156.195 | [open address](#)

Instance state
Running

Private IPv4 addresses
172.31.1.43

Public DNS
ec2-13-203-156-195.ap-south-1.compute.amazonaws.com | [open address](#)

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(ii) Create an Application Load Balancer (ALB) to distribute traffic across these instances.

Create Target Group

tg-webapp-22bbs0155

The screenshot shows the 'Create target group' page in the AWS Management Console. The 'Target group name' field is filled with 'tg-webapp-22bbs0155'. The 'Protocol' is set to 'HTTP' and the 'Port' is '80'. The 'IP address type' is set to 'IPv4'. The 'VPC' is set to 'vpc-010499e03cdd3d905'. The 'Protocol version' is set to 'HTTP1'. The page includes instructions for each field and a 'Create VPC' button.

Target group name: tg-webapp-22bbs0155

Protocol: HTTP

Port: 80

IP address type: IPv4

VPC: vpc-010499e03cdd3d905

Protocol version: HTTP1

The screenshot shows the 'Register targets' page in the AWS Management Console. It displays a table of available instances with two instances selected. The 'Ports for the selected instances' field is set to '80'. The 'Include as pending below' button is visible.

Register targets

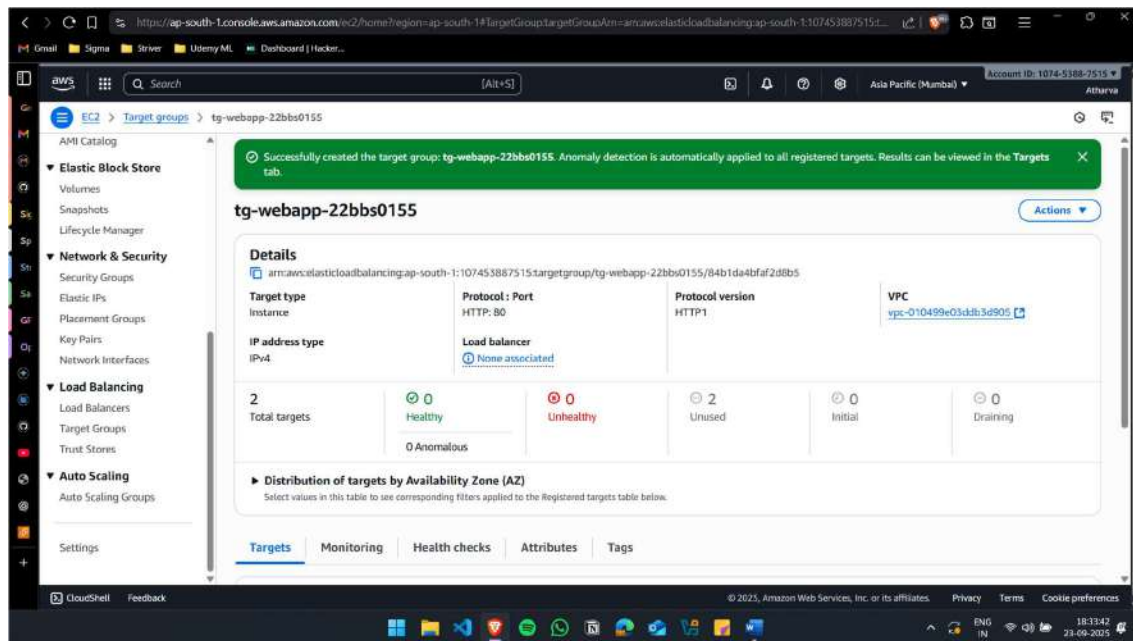
Available instances (2/2)

Instance ID	Name	State	Security groups	Zone
i-0def76871d9736448	web-b-22bbs0155	Running	web-22bbs0155	ap-sou1
i-01740e35e38cd0671	web-a-22bbs0155	Running	web-22bbs0155	ap-sou1

2 selected

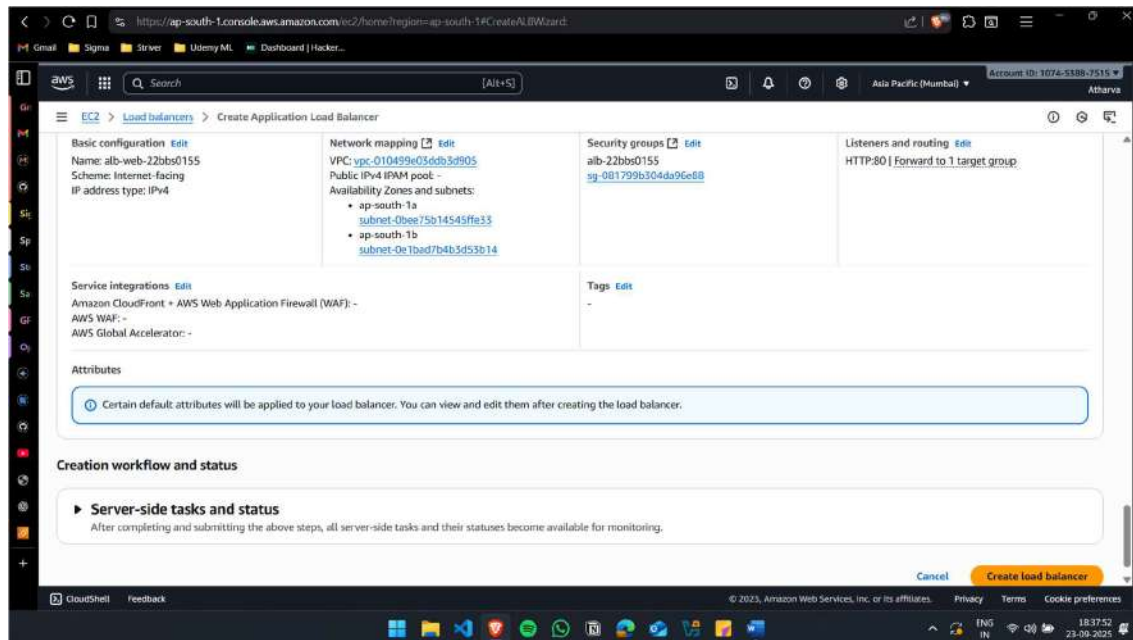
Ports for the selected instances: 80

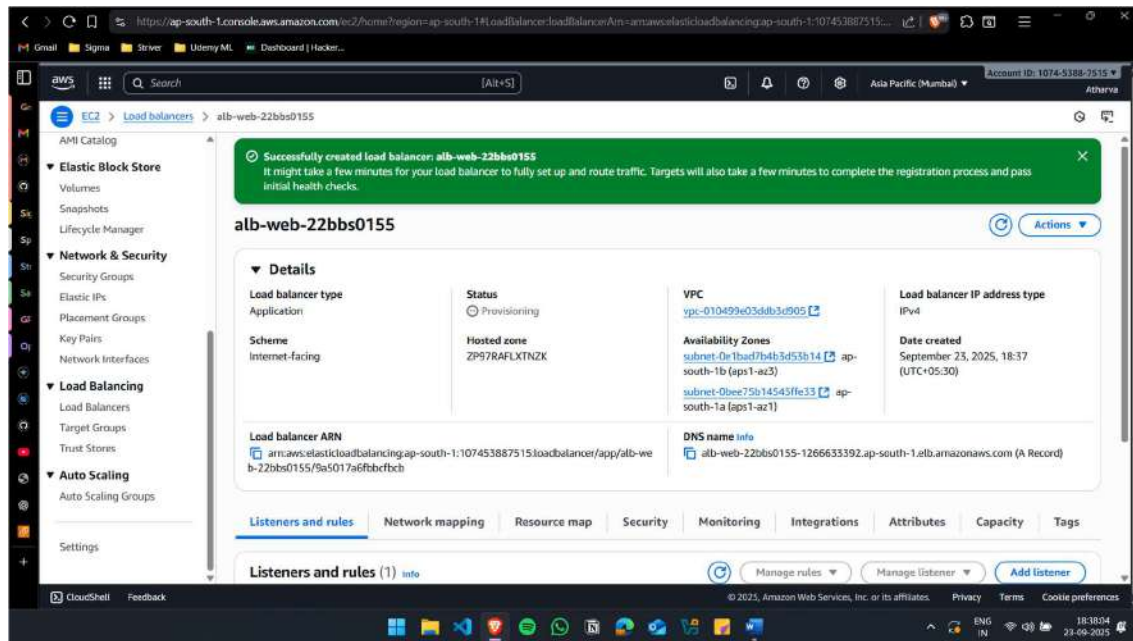
Include as pending below



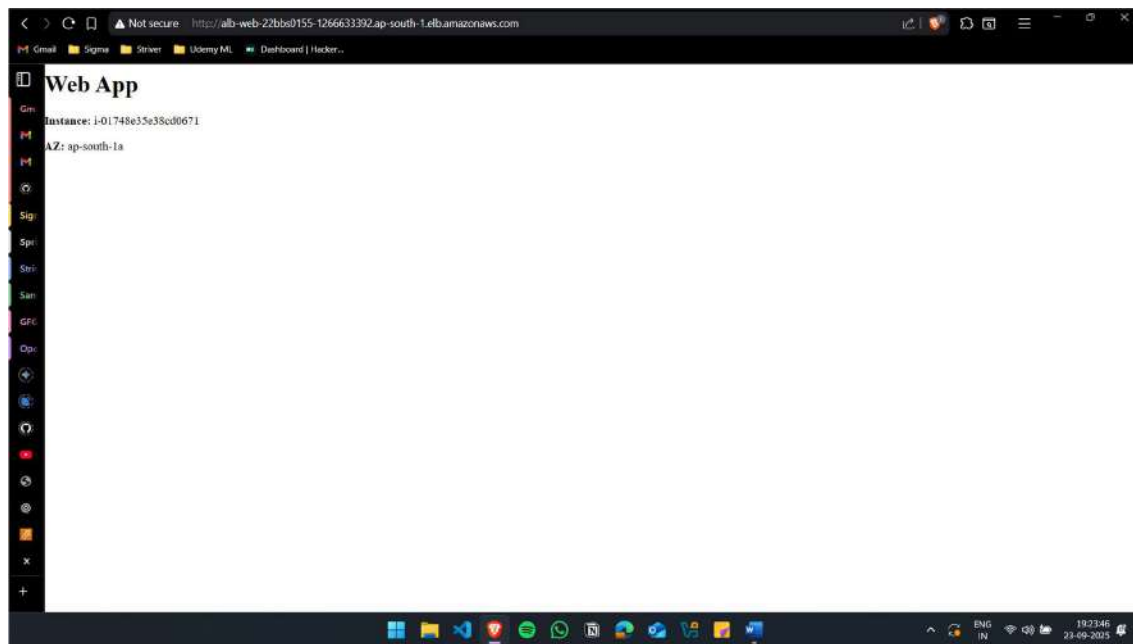
Create the Automatic Load Balancer

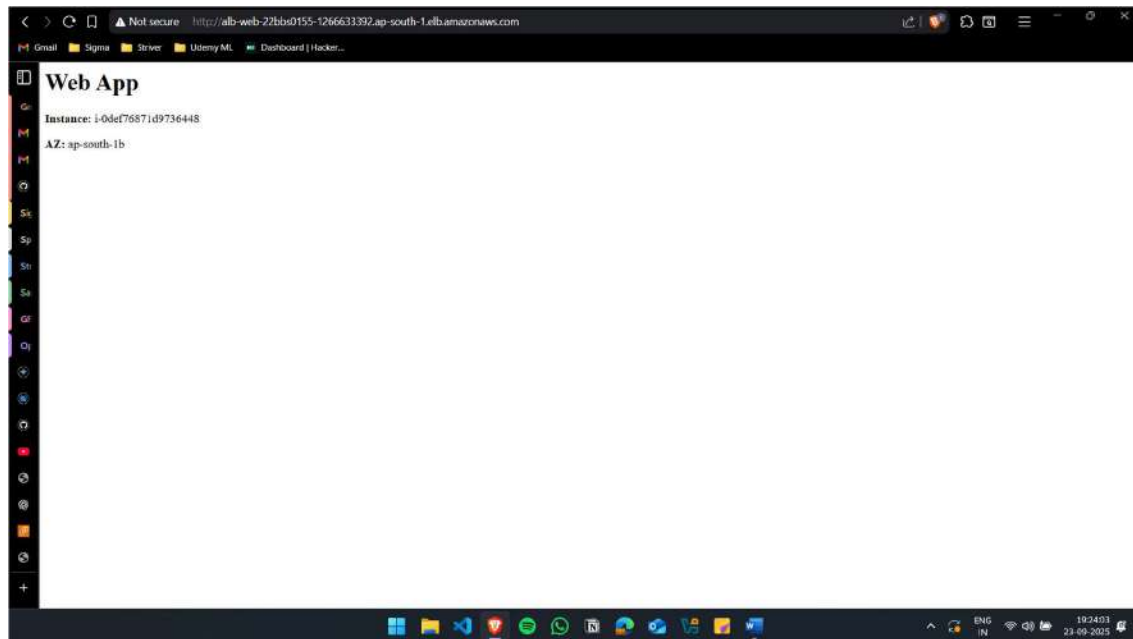
alb-web-22bbs0155





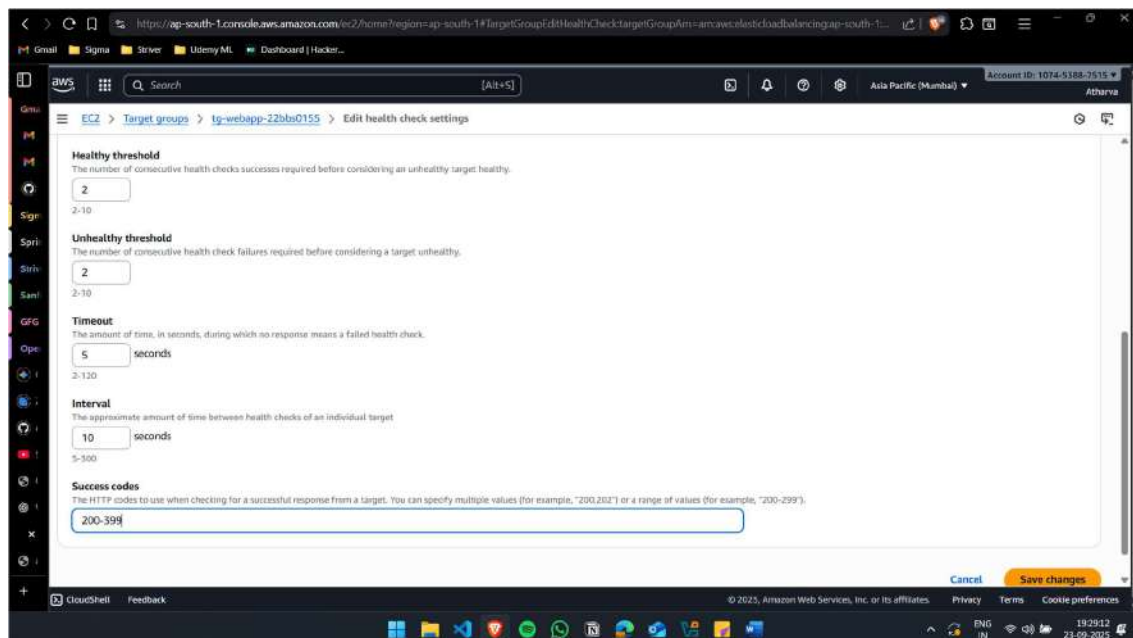
Accessing the Files from the ALB DNS (After refreshing)

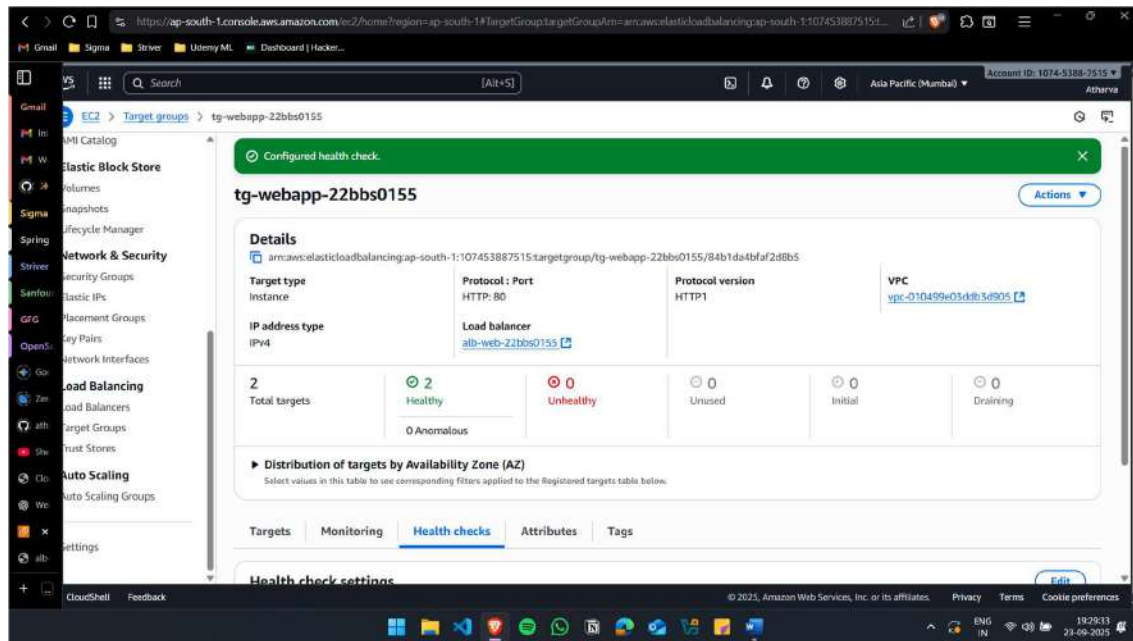




(iii) Configure health checks so that faulty instances are automatically removed from load balancing.

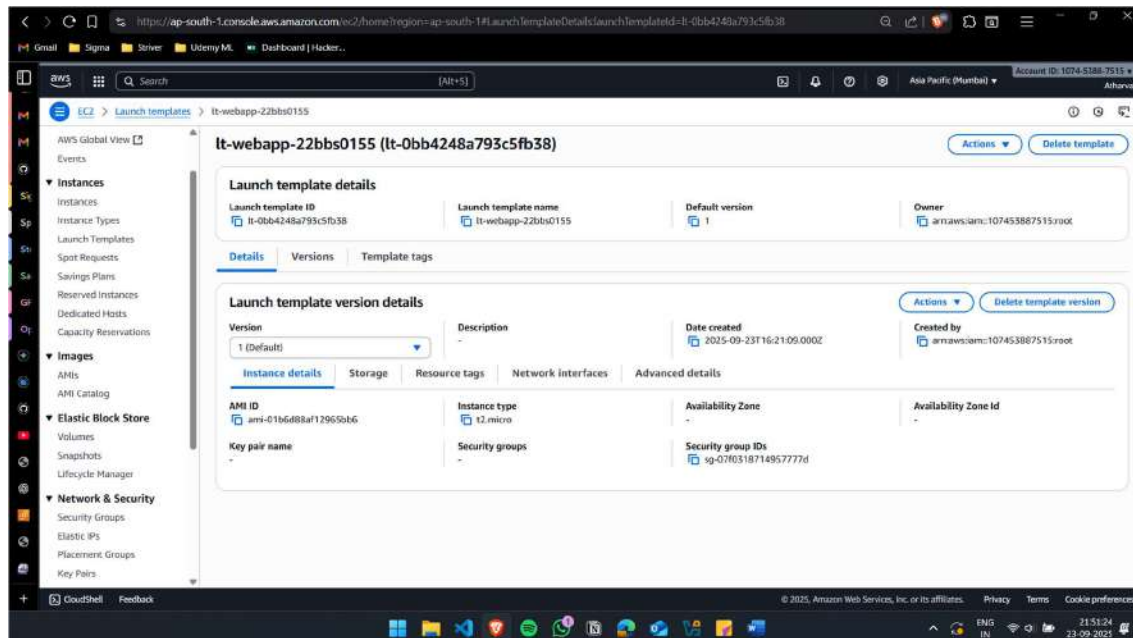
Edit Health Check Settings

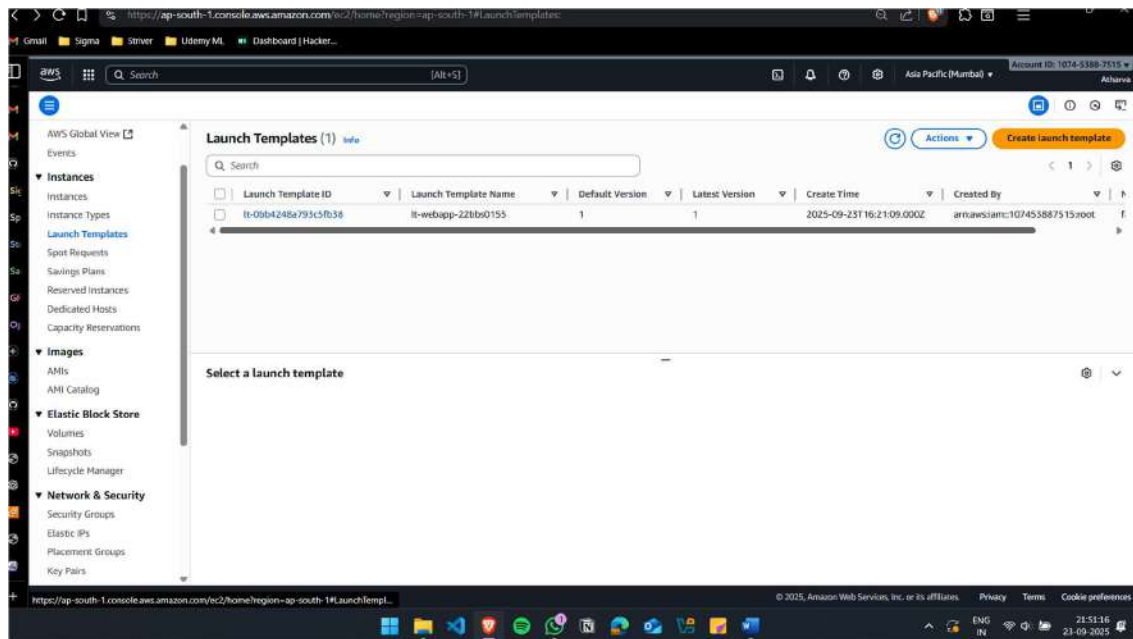




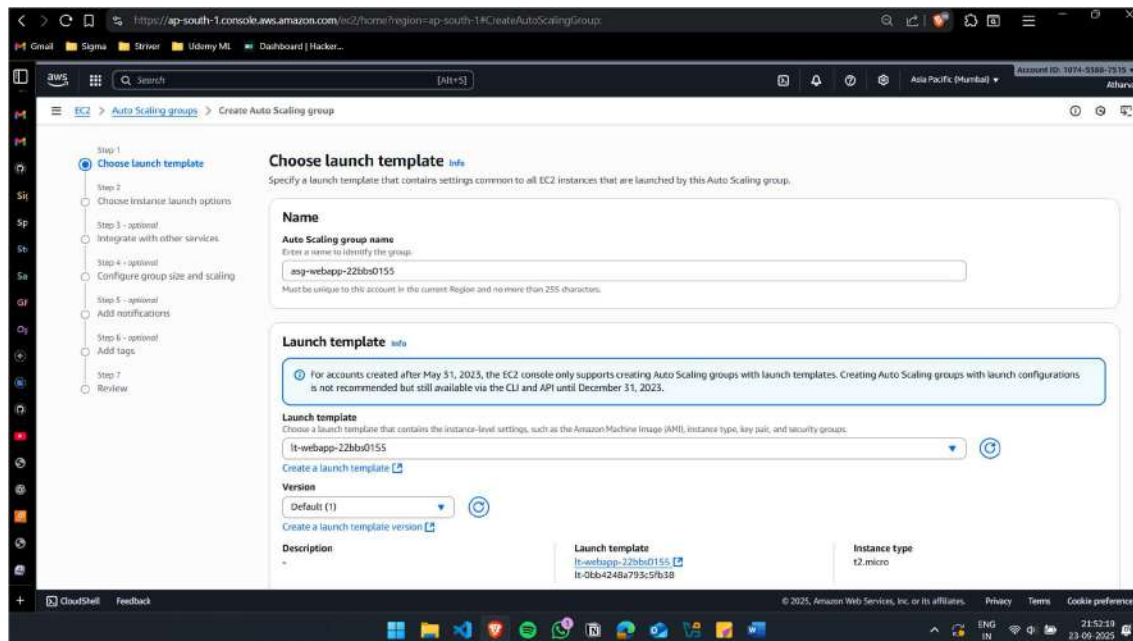
(iv) Enable Auto Scaling to add/remove instances based on traffic demand.

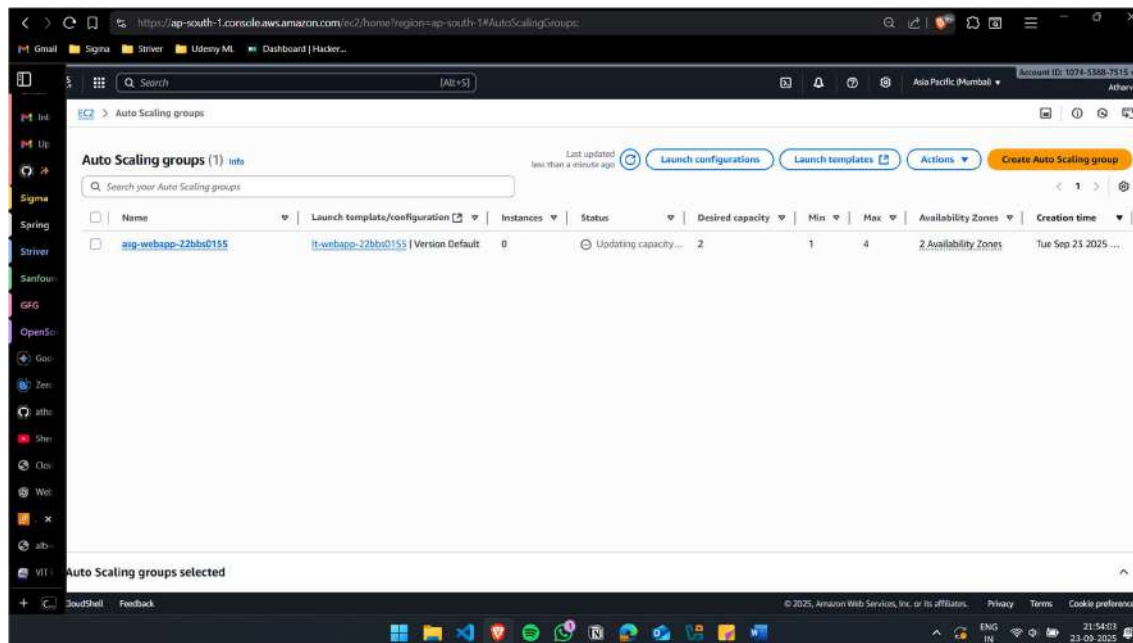
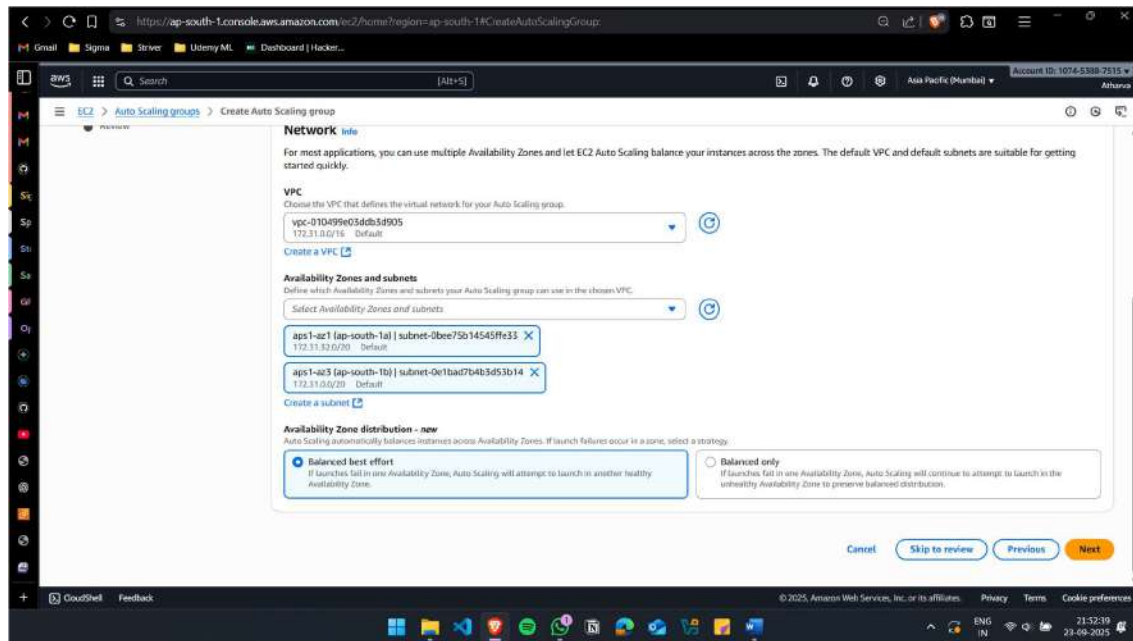
Create Launch Template

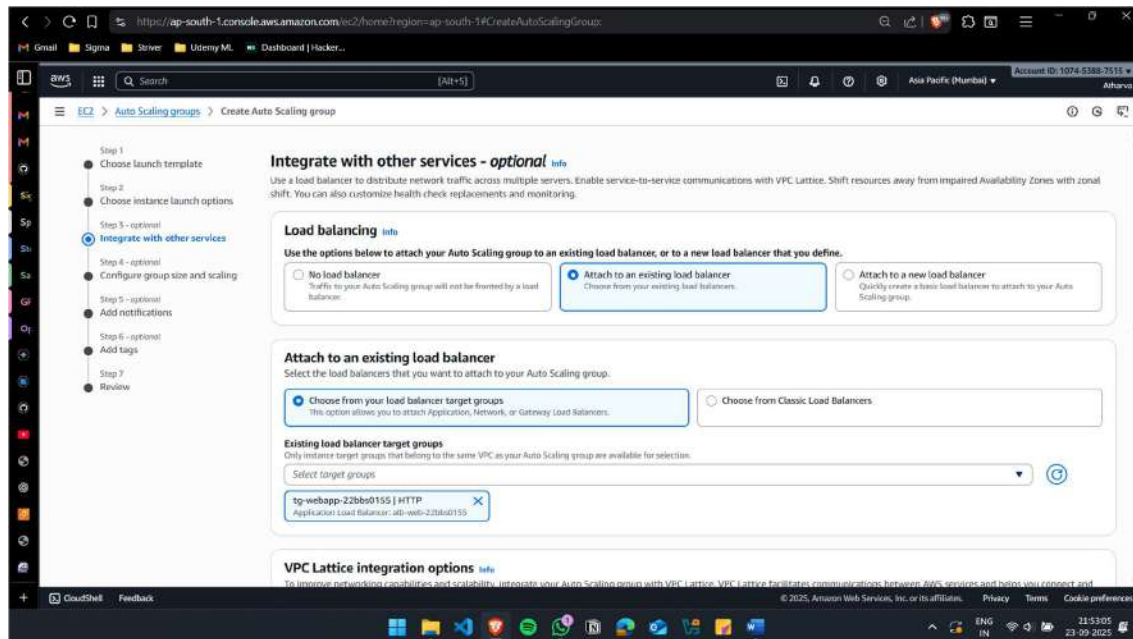




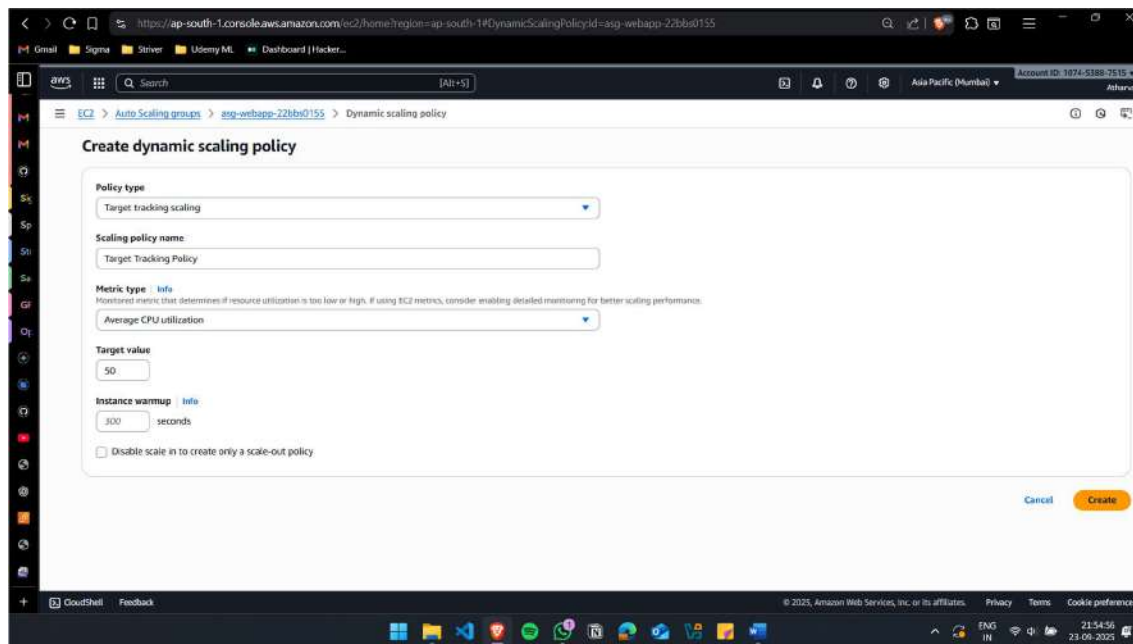
Create Auto Scaling Groups

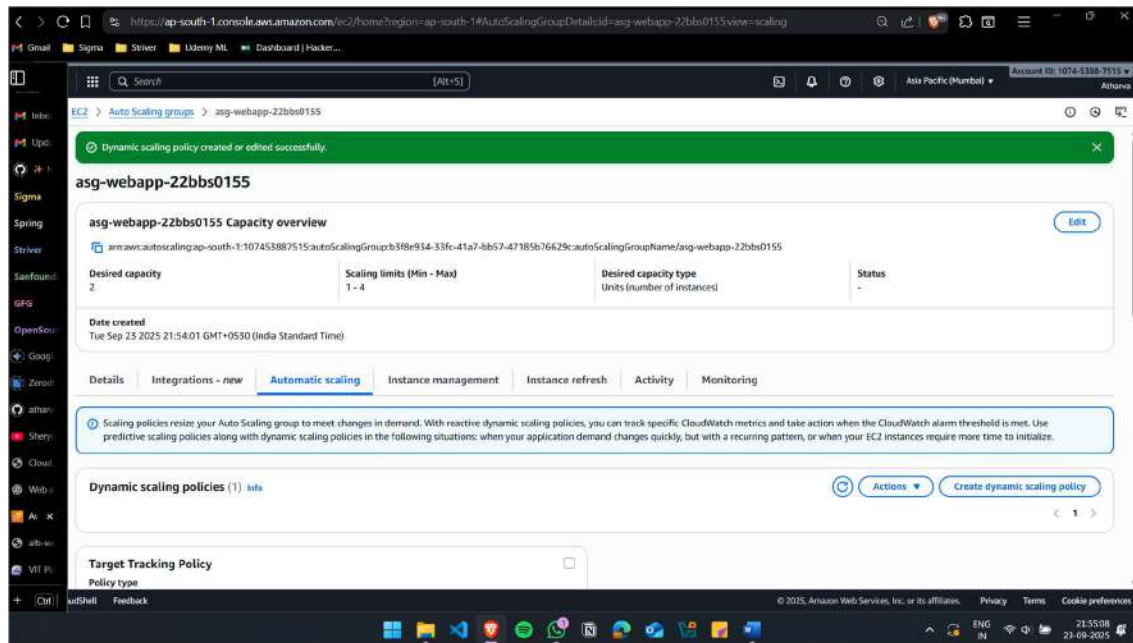






Create Scaling Policy

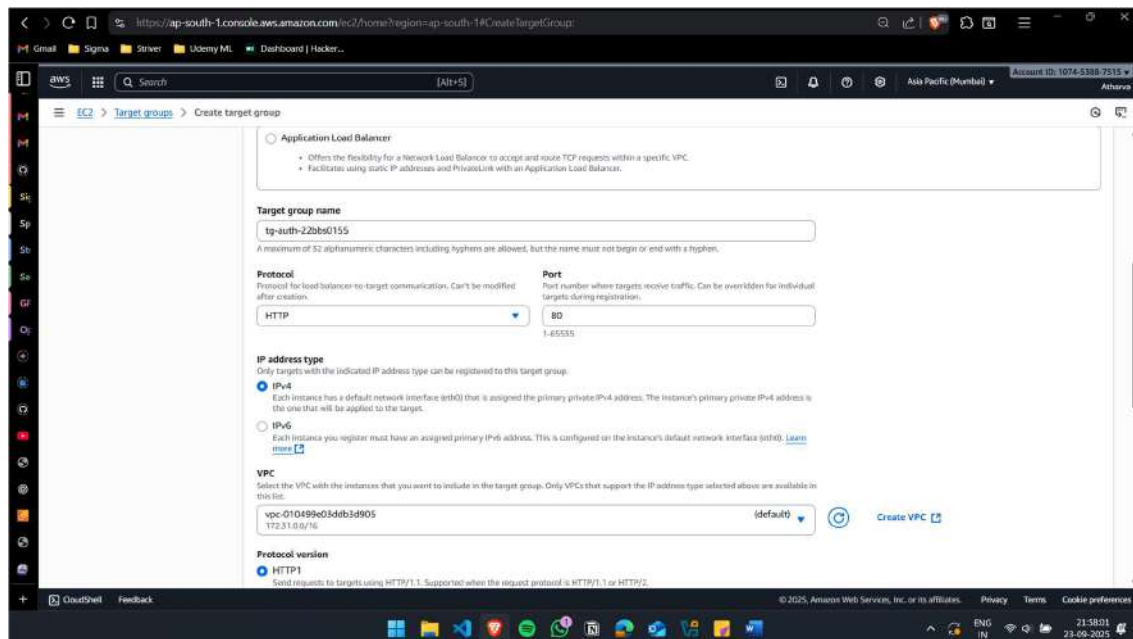


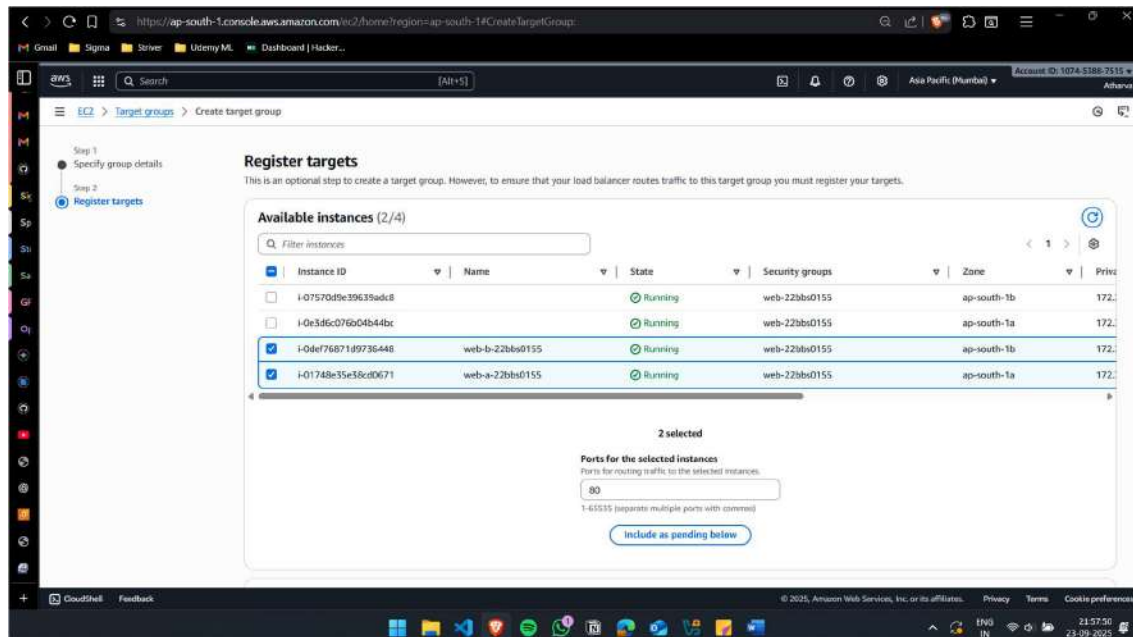


(v) Configure path-based routing: /auth requests go to the authentication service, /order requests go to the order processing service. Integrate the Load Balancer with Route 53 so that global users are routed to the nearest AWS region.

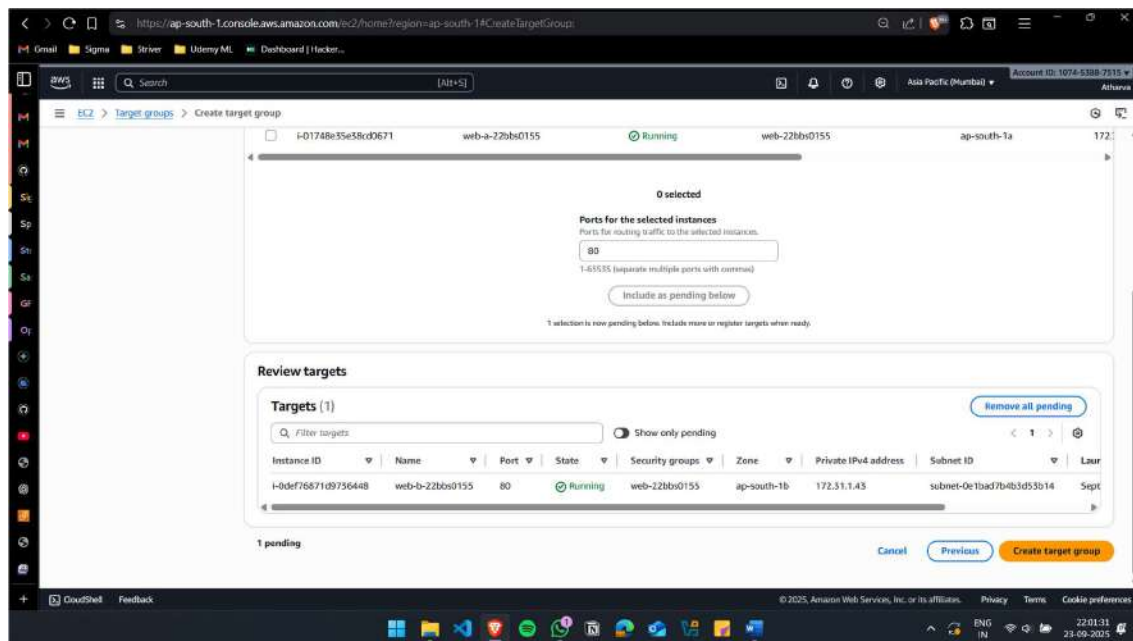
For this, instead of creating additional instances, I'm assuming web-a as Auth and web-2 as order.

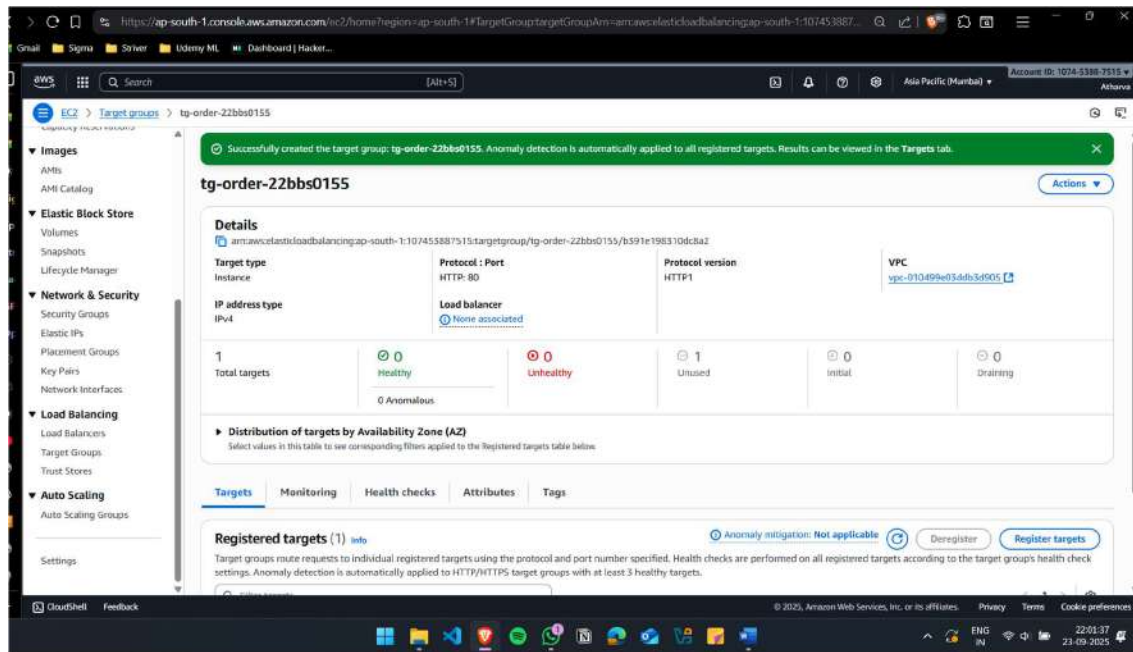
Create new Target Group for Auth(Web-a)



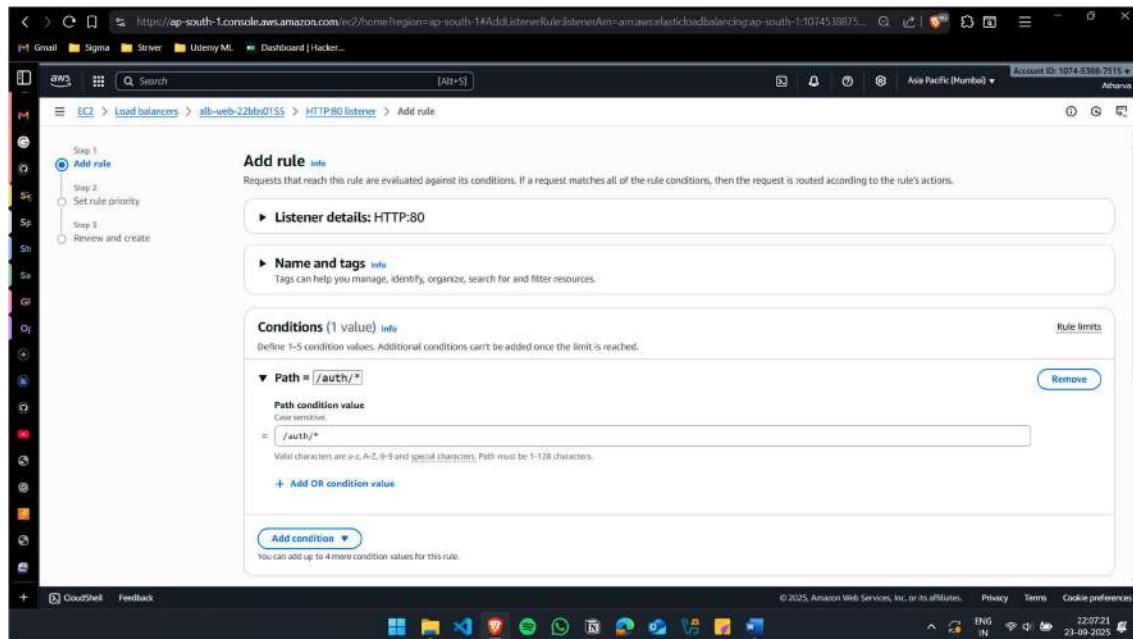


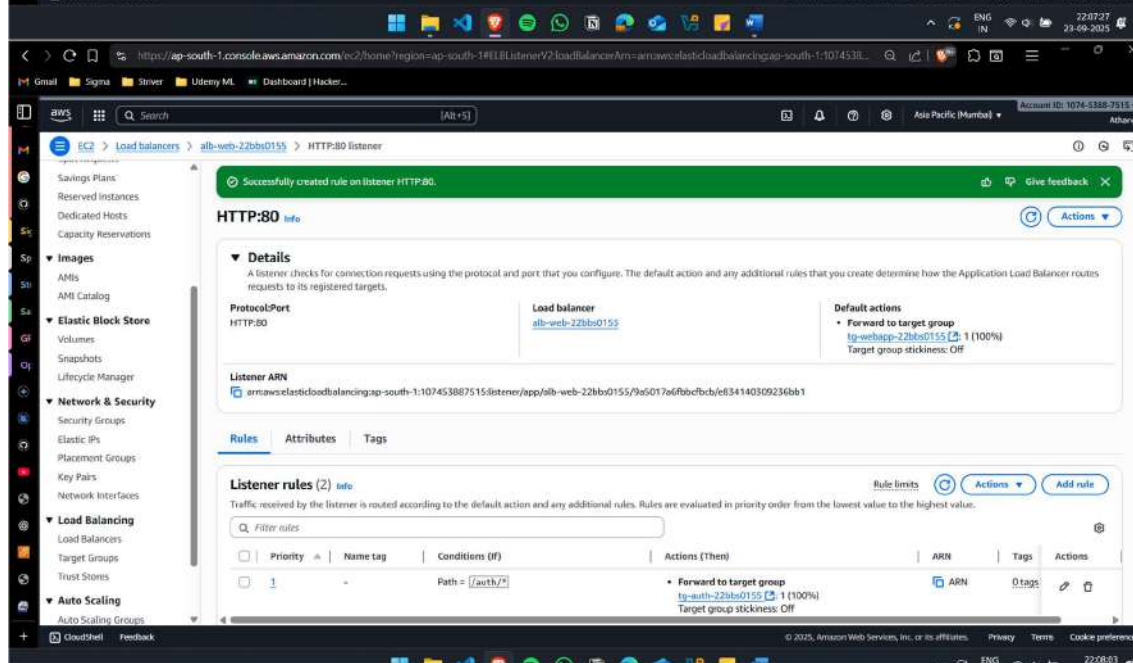
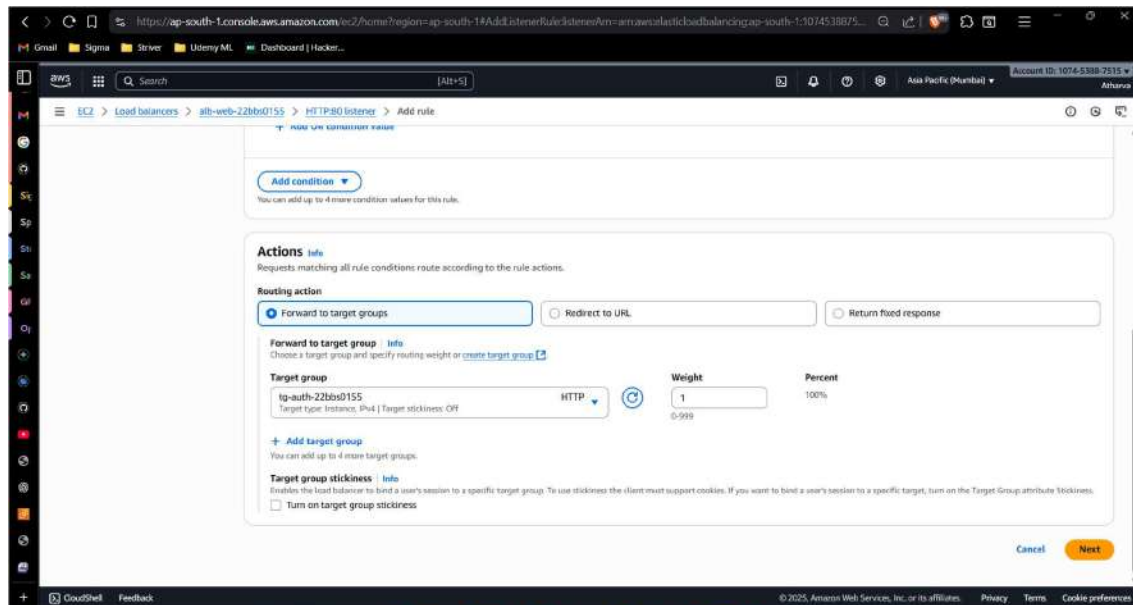
Create new Target Group for Order(Web-b)

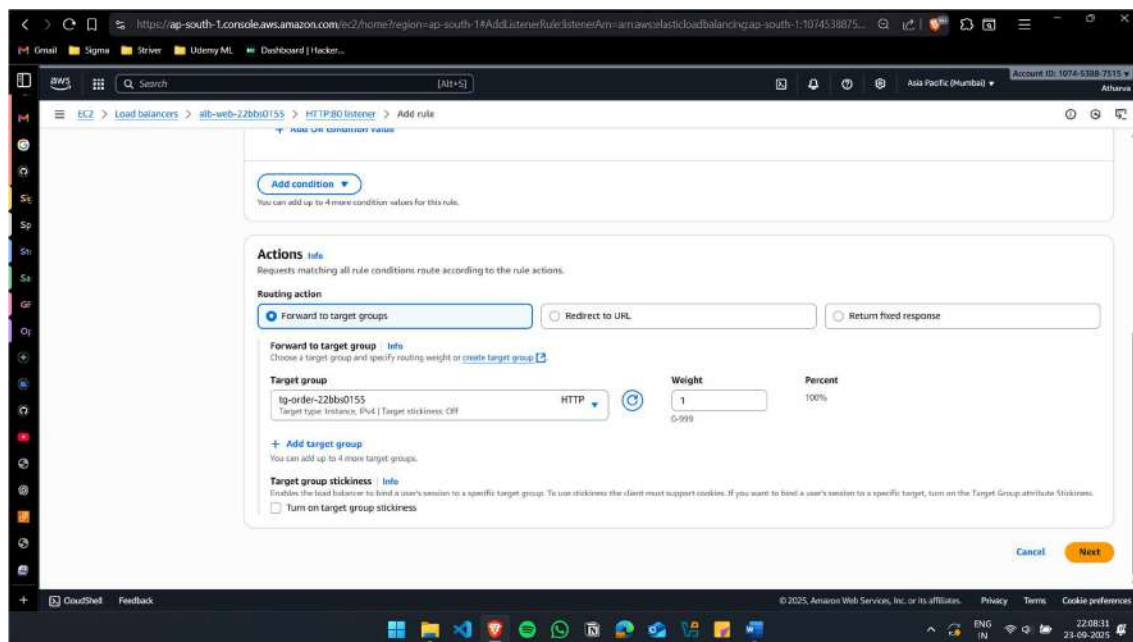
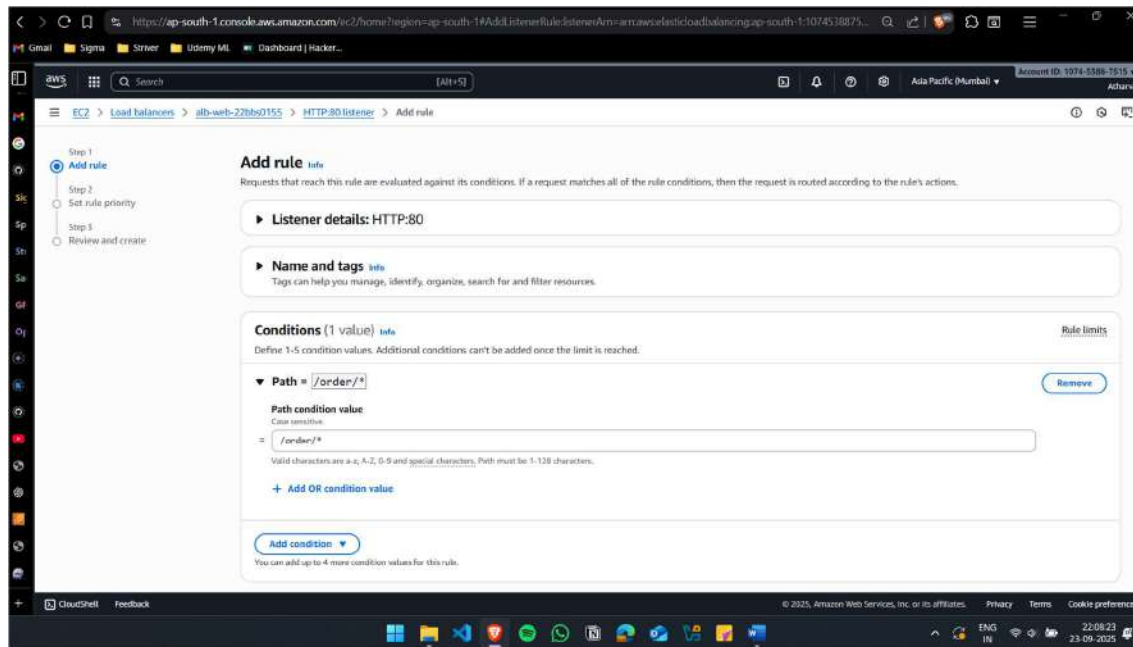


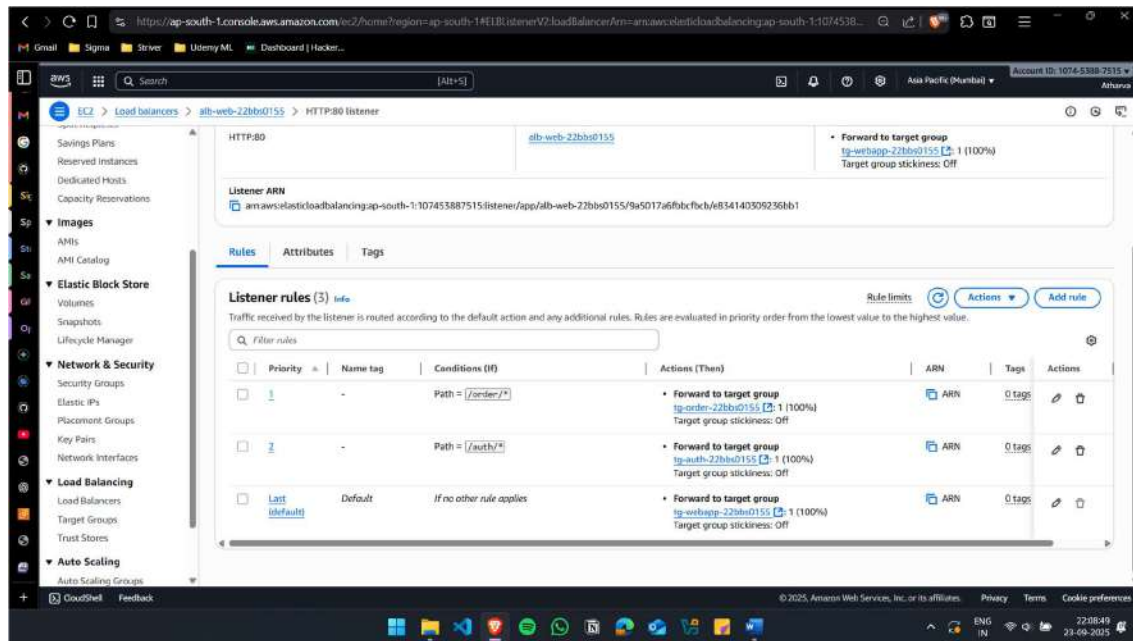


Add the rules to the ALB

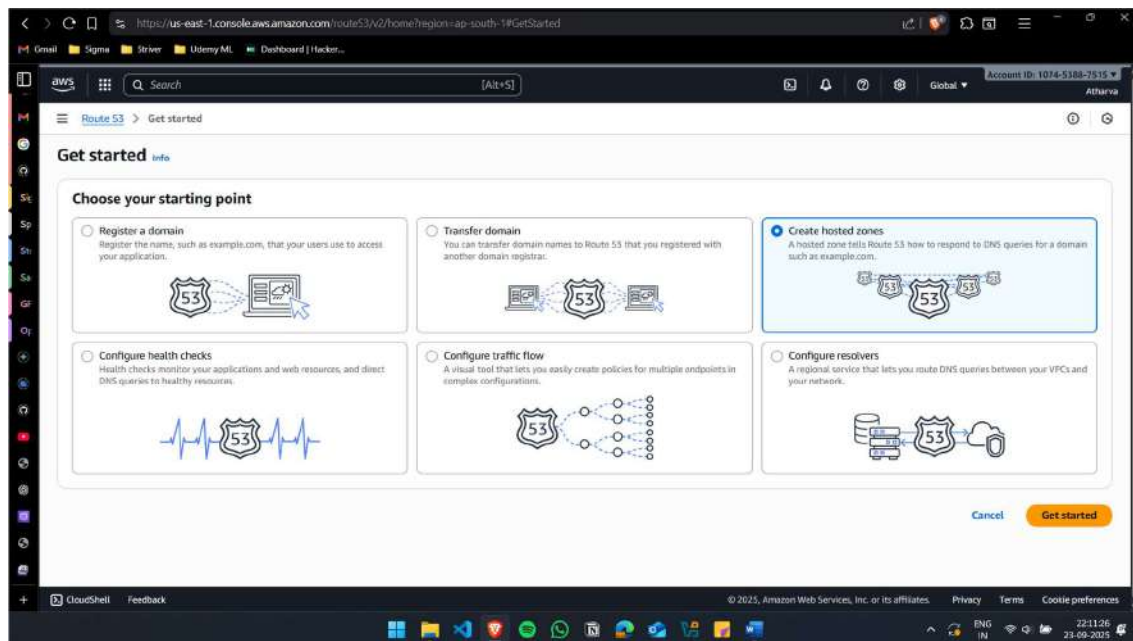


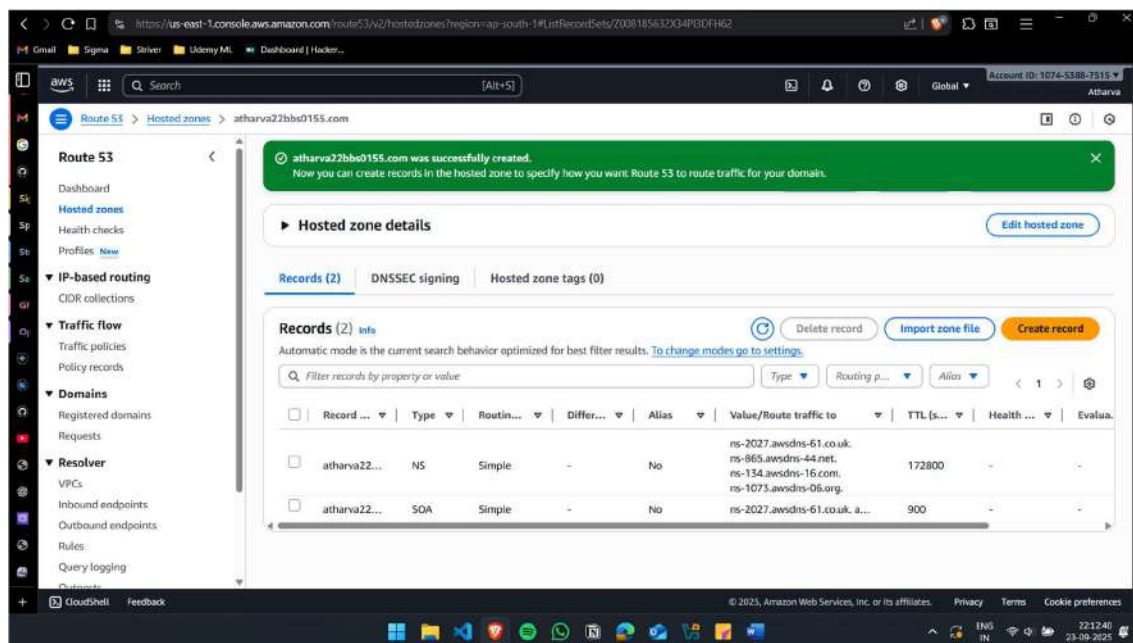
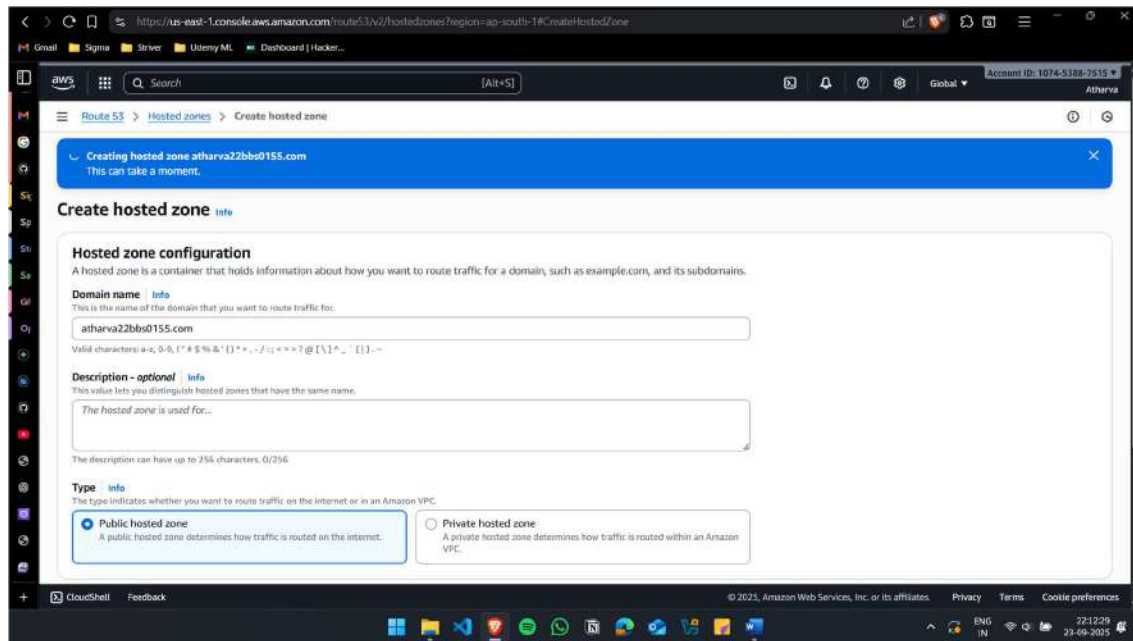






Route to the nearest hosted region





Create 2 records(mumbai and singapore)

https://us-east-1.console.aws.amazon.com/route53/v2/hostedzones?region=ap-south-1&CreateRecordSet/Z008185632X34P3DFH62

Record 1

Record name: .atharva22bbs0155.com

Record type: A - Routes traffic to an IPv4 address and some AWS resources

Alias: ☒ Alias

Route traffic to: Alias to Application and Classic Load Balancer

Asia Pacific (Mumbai)

Routing policy: Latency

Region: Asia Pacific (Mumbai)

Health check ID - optional:

Evaluate target health: ☒ Yes

Record ID:

CloudShell Feedback

https://us-east-1.console.aws.amazon.com/route53/v2/hostedzones?region=ap-south-1&CreateRecordSet/Z006185632X34P3DFH62

Create record

Record name: .atharva22bbs0155.com

Record type: A - Routes traffic to an IPv4 address and some AWS resources

Alias: ☒ Alias

Route traffic to: Alias to Application and Classic Load Balancer

Asia Pacific (Singapore)

Routing policy: Latency

Region: Asia Pacific (Singapore)

Health check ID - optional:

Evaluate target health: ☒ Yes

Record ID:

Add another record

CloudShell Feedback

https://us-east-1.console.aws.amazon.com/route53/v2/hostedzones?region=ap-south-1&idRecordSets/2008185612G34P3D1H62

Search [Alt+S]

Account ID: 1074-8388-7515 Atharva

Route 53 > Hosted zones > atharva22bbs0155.com

Records for atharva22bbs0155.com were successfully created. Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Use "View status" button to check propagation status. [View status](#)

Records (4) DNSSEC signing Hosted zone tags (0)

Records (4) info [Delete record](#) [Import zone file](#) [Create record](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias < 1 >

☐	Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...	Evalu...
☐	atharva22...	NS	Simple	-	No	ns-2027.awsdns-61.co.uk. ns-065.awsdns-44.net. ns-154.awsdns-16.com. ns-1073.awsdns-06.org	172800	-	-
☐	atharva22...	SOA	Simple	-	No	ns-2027.awsdns-61.co.uk. a...	900	-	-
☐	www.atha...	A	Latency	Asia Paci...	Yes	dualstack.alb-web-22bbs015...	-	-	Yes
☐	www.atha...	A	Latency	Asia Paci...	Yes	dualstack.alb-web-22bbs015...	-	-	Yes